

Cells With Odd Values in Matrix

Statement: Ex: $m=2$ $n=3$ indices $[[0,1], [1,1]]$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

indices
 $[0,1]$

increment
1st row
 $[0,1]$
row column

increment
1st column

indices $[1,1]$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix}$$

increment row 1

increment column 1

Now check how many odd numbers are

present in final matrix $\begin{bmatrix} 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix} \Rightarrow 6$ odd numbers

Output = 6

Approach with 2 explanation :-

→ First we need a matrix of $m \times n$ of 0's ($\text{array}[m][n]$) $\downarrow$
row $\downarrow$
column

2) Use for loop for indices (array) ($i < \text{indices.length}$)
 ↳ create variable row for 0th index
 ↳ create variable col for 1th index } why - this?

↓ Explanation

The question say $\text{indices}[\text{row}, \text{col}]$ that means only 2 elements & (0, 1) index would be present in each array.

Ex: $\text{indices}[\text{row}, \text{col}]$

$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

So 2 elements in each row.

3) increment row by 1 so use for-loop like this:

$\text{row is always at 0-th index}$

$\text{ans}[\text{row}][i]++;$

$\downarrow$ $\downarrow$
fixed increment

it will increase the row by 1 but for loop $j < n$, not m (why?)

Explanation

for ex: $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ $m=2$ (row)
 $n=3$ (col)

1st we have to increase row like this

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

so if we write $j < m$

so m is 2

only it will not increment

whole row but if we write n so $n=3$ that's why.

4) Use for loop for column and $k < m$ not n b/c

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ increment 2 element & $m=2$ & $n=3$ so m would be

→ fixed
ans [1<][col]++;

↳ column is always at
1st index

4) check whether the find matrix has
odd elements & then return how many
odd numbers are found in find matrix
ans:

```
so use for-loop {  
    for loop {  
        if (ans[i][j] % 2 != 0) {  
            count++;  
        }  
    }  
}
```